SATPURA ANALYTICAL LABORATORIES

Chemical & Microbiological Food Testing Division
ISO/IEC 17025 Compliant Testing Facility (Fictional Demonstration)

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CERTIFICATE OF QUALITY ANALYSIS REPORT

1. DOCUMENT & SAMPLE IDENTIFICATION

Report Reference:	SIL/LQR/2026/09-0482	**Sample Received:**	08-Sep-2026
Issue Date:	11-Sep-2026	**Analysis Commenced:**	08-Sep-2026
Client Name:	HoneyChain Procurement Operations	**Analysis Completed:**	10-Sep-2026
Client Reference:	Traceability Demonstration Project	**Sample Condition:**	Intact, Sealed Glass Jar (500g)

2. SAMPLE DESCRIPTION & ORIGIN DETAILS

Sample Name:	Raw Unprocessed Honey	**Batch ID / Lot No.:**	**HC-SEP-REJ**
Declared Origin:	Central India Forest Belt	**Sample Code (Lab):**	SAL-2026-0908-012
Purpose of Test:	Comprehensive Quality & Purity Assessment	**Storage Condition:**	Ambient (20°C - 25°C)

3. LABORATORY ANALYTICAL RESULTS

TEST PARAMETER	TEST METHOD / REFERENCE	UNITS	QUALITY SPECIFICATION / LIMIT	OBSERVED RESULT	STATUS
Appearance	IS 4941 : 2019 / Visual	-	Viscous liquid, free from foreign matter	Viscous amber liquid	**PASS**
Odour / Aroma	IS 4941 : 2019 / Organoleptic	-	Characteristic honey aroma, no off-odours	Characteristic, slight thermal note	**PASS**
Moisture Content	AOAC 969.38 (Refractometry)	% (m/m)	Max 20.0	17.8	**PASS**
Hydroxymethylfurfural (HMF)	AOAC 980.23 / HPLC-UV	mg/kg	Max 40.0 (Tropical Max 80.0)	**98.4**	**FAIL**
Diastase Activity	AOAC 958.09 (Schade Method)	Gothe Unit	Min 8.0	11.2	**PASS**
Reducing Sugars (Fructose + Glucose)	IS 4941 : 2019 (Lane & Eynon)	% (m/m)	Min 65.0	68.5	**PASS**
Sucrose Content	IS 4941 : 2019 (Titrimetric)	% (m/m)	Max 5.0	2.4	**PASS**
Free Acidity	AOAC 962.19 (Titrimetry)	meq/kg	Max 50.0	32.0	**PASS**
Electrical Conductivity	IHC Method 2009	mS/cm	Max 0.80	0.48	**PASS**

TEST PARAMETER	TEST METHOD / REFERENCE	UNITS	QUALITY SPECIFICATION / LIMIT	OBSERVED RESULT	STATUS
C4 Sugar Adulteration (EA-IRMS)	AOAC 998.12	%	Max 7.0 (Negative)	-25.8 ‰ (< 2.0% C4)	**PASS**
Foreign Sugar Screening (SMR/TMR)	LC-HRMS Screen	-	Absent	Not Detected	**PASS**

- * Limits referenced as per standard quality benchmarks for commercial honey specifications.
- * Test parameters evaluated under controlled environment at 20°C ± 2°C.

4. SUMMARY EVALUATION & FINAL CONCLUSION

NON-CONFORMING / REJECTED

Evaluation Summary:

The submitted honey sample (Batch ID: **HC-SEP-REJ**) was subjected to standard physical, chemical, and purity parameter evaluations. All composition metrics including moisture, reducing sugars, sucrose, free acidity, electrical conductivity, and foreign sugar screening conform strictly to acceptable quality specifications.

Primary Non-Conformity:

The sample exhibits a severe elevation in Hydroxymethylfurfural (HMF) level at **98.4 mg/kg**, which substantially exceeds the standard maximum permissible threshold (40.0 mg/kg) as well as the extended tropical limit threshold (80.0 mg/kg). High HMF values typically indicate excessive thermal treatment, improper processing temperatures, or prolonged storage under elevated ambient conditions.

Final Conclusion Statement:

The tested sample DOES NOT CONFORM to the required quality and freshness specifications due to excessive HMF accumulation. Batch HC-SEP-REJ is declared UNSUITABLE FOR RELEASE and must be quarantined pending appropriate commercial disposition or recall procedures.

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